

2.2 Programming fundamentals

Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 02: Computational thinking, algorithms and programming — 80 marks, 1 hour 30 minutes
What OCR says the component covers	computational thinking: algorithms, programming techniques, producing robust programs, computational logic and translators
This unit	topic 2.2 Programming fundamentals
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: 1.1 Systems architecture, 1.2 Memory and storage, 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms

After it: 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Variable store (pf-101)	How does a program store data, and how does it decide which line to run next?	14	21
2	Type foundry (pf-102)	Why does a program need to know what type of data it is working with?	14	19
3	Filing room (pf-103)	How does a program keep hold of data after it has stopped running?	14	18
4	Record vault (pf-104)	How does a program store a set of related data, and how does it search it?	14	19
5	Array yard (pf-105)	How does one name hold many values, and how is code broken into reusable pieces?	14	19
6	Chance machine (pf-106)	How does a program produce a number that nobody can predict?	14	19
7	Revision HQ 2.2 (pf-107)	Which parts of 2.2 am I secure on, and which do I need to revise before the test?	15	20

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
8	End-of-topic test pf-l08-test	assesses the whole unit	0	0

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1

Vocabulary introduced: Assignment, Count-controlled iteration, Casting, Concatenation, File handling, Append, Record, Wildcard, Zero-indexing, Sub-program, Pseudo-random, Seed, Iteration

What a pupil does, not only reads: match, mcq, multi, order, sort

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in `build/coverage.json`. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know what variables and constants are, and how a value is assigned to each pf-l01-o1	pf-l01	yes	yes	yes	Full
understand how input and output move data in and out of a program pf-l01-o2	pf-l01	yes	yes	yes	Full
know what sequence, selection and iteration each do to the order a program runs in pf-l01-o3	pf-l01	yes	yes	yes	Full
know the data types integer, real, Boolean, character and string pf-l02-o1	pf-l02	yes	yes	yes	Full
understand why casting is needed and how a value is cast from one type to another pf-l02-o2	pf-l02	yes	yes	yes	Full
know how the arithmetic, comparison and Boolean operators are used pf-l02-o3	pf-l02	yes	yes	yes	Full
know why a program needs to store data in a file pf-l03-o1	pf-l03	yes	yes	yes	Full
understand how a file is opened, read from, written to and closed pf-l03-o2	pf-l03	yes	yes	yes	Full
know the difference between writing to a file and appending to it pf-l03-o3	pf-l03	yes	yes	yes	Full
know what is meant by a record, a field and a table of records pf-l04-o1	pf-l04	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand how SELECT, FROM and WHERE are used to search for data pf-104-o2	pf-l04	yes	yes	yes	Full
know how AND, OR, NOT and LIKE narrow a search down pf-104-o3	pf-l04	yes	yes	yes	Full
know how one- and two-dimensional arrays are declared, read and changed pf-105-o1	pf-l05	yes	yes	yes	Full
understand why arrays are zero-indexed and how a loop iterates over one pf-105-o2	pf-l05	yes	yes	yes	Full
know the difference between a procedure and a function, and between local and global variables pf-105-o3	pf-l05	yes	yes	yes	Full
know what random numbers are used for in a program pf-106-o1	pf-l06	yes	yes	yes	Full
understand how a random whole number is generated within a given range pf-106-o2	pf-l06	yes	yes	yes	Full
know that a computer produces pseudo-random numbers from an algorithm and a seed pf-106-o3	pf-l06	yes	yes	yes	Full
know the whole of topic 2.2 well enough to answer without notes pf-107-o1	pf-l07	—	—	—	Not covered
understand the mistakes that most often lose marks in programming questions pf-107-o2	pf-l07	—	—	—	Not covered
know which lessons are secure and which still need revision before the test pf-107-o3	pf-l07	—	—	—	Not covered

18 full · **0 partial** · **3 not covered**, of 21 outcomes.

Gaps, assumptions and overlap

None found by the trace.

Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- pf-101-x3 A program asks for a password and keeps asking until the correct one is entered. Write an algorithm to do this, using the OCR Exam Reference Language or a language you know.
- pf-102-x1 Identify the most suitable data type for each of these values: 17, 3.5, True, "Year 10".
- pf-108-test-x2 -
- pf-108-test-x4 -
- pf-108-test-x6 -
- pf-108-test-x7 String manipulation
- pf-108-test-x8 -

- pf-108-test-x10 -
- pf-108-test-x11 Records and SQL
- pf-108-test-x12 -
- pf-108-test-x14 -
- pf-108-test-x15 Procedures, functions, parameters and scope
- pf-108-test-x16 -
- pf-108-test-x18 -

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